

Qinnan Jiang

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PROFILE

Electrical and Electronic Engineering undergraduate at the University of Bristol with an average of 83% (First-Class), ranked top 1% of cohort. Hands-on robotics engineer working across mobile navigation, robot manipulation, visual perception, and reinforcement learning. Seeking graduate research in robotics, robot perception, robot learning and embodied intelligence.

EDUCATION

University of Bristol — Bristol, United Kingdom 2024 – Expected 2027

BEng (Hons) Electrical and Electronic Engineering | On track for First Class (UK threshold = 70) | Average: 83%

Year 2 (2025/26) — Average 83%

Module	Credits	Mark
Electromagnetics and Semiconductor Devices	20	99
Signals and Communications	20	95
Engineering Mathematics 2	20	82
Electrical and Electronic Engineering Practice	40	76
Electrical Energy Conversion and Supply	20	70

PROFESSIONAL EXPERIENCE

Autonomous Robotics Systems Intern — Beijing SunstepVision Co., Ltd. Jun 2026 – Sep 2026

- Owned end-to-end development across three robot systems: autonomous mobile navigation, dual-arm vision-based manipulation, and real-time dual-motor synchronous control, from hardware integration to field deployment
- Integrated perception (YOLOv11-Seg, FoundationPose), planning (Nav2, MoveIt 2), and control (ros2_control, CAN bus) stacks on real hardware platforms operating in indoor and outdoor environments
- Designed zero-torque gravity-compensation controllers to enable kinesthetic teaching and VLA data collection

Robot Learning R&D Intern — Shanghai Immersion Information Technology Co., Ltd. May 2025 – Sep 2025

- Built a complete reinforcement learning training pipeline for robot locomotion in Isaac Gym, from SolidWorks CAD modeling and URDF export to PPO policy training and Sim2Real deployment
- Evaluated NVIDIA Omniverse, Isaac Sim and Isaac Lab toolchains for simulation-based robot learning workflows
- Conducted multi-sensor testing (LiDAR, mmWave radar, RGB-D, IMU), 5G+GPS communication testing on RK3588

PROJECTS

Dual 7-DoF Robot Arm Jul 2026 – Sep 2026

Completed during internship at SunstepVision | github.com/nanj-robotics/RoboMantis

C++ / Python · ROS 2 Humble · MoveIt 2 · ros2_control · YOLOv11-Seg · FoundationPose · CAN bus · Pinocchio

- Designed and built a custom dual 7-DoF manipulator end to end: SolidWorks-to-URDF model, independent dual-arm ros2_control hardware interfaces, MoveIt2 dual-arm planning configuration, and grasp-execution node
- Built the full perception-to-planning chain: YOLOv11-Seg instance segmentation → FoundationPose 6D pose estimation → eye-to-hand calibration → camera-to-base transform → MoveIt 2 planning, achieving 90% grasp success over 20 trials
- Collected and labelled a 100-image RGB-D dataset and trained YOLOv11-Seg model; implemented gravity-compensated zero-torque controllers via Pinocchio RNEA with adaptive damping for kinesthetic teaching and VLA data collection

Dual-Motor Synchronous Control Aug 2026

Completed during internship at SunstepVision | github.com/nanj-robotics/dual_motor_sync

C++ · ROS 2 Humble · CAN bus · trajectory planning · real-time control

- Authored a ROS 2 C++ package synchronizing two mirrored motors over CAN using time-parameterised cosine trajectories and 500 Hz back-to-back frame transmission in MIT PD-torque mode
- Achieved synchronization error down to 0.01° RMS

Autonomous Mobile Robot Jun 2026 – Jul 2026

Completed during internship at SunstepVision | github.com/nanj-robotics/geni_rover

Python / C++ · ROS 2 Humble · Nav2 · SLAM/AMCL · EKF · RTK-GNSS · nvblox · NVIDIA Jetson AGX Orin

- Developed a four-wheel differential-drive robot supporting both indoor SLAM navigation and outdoor RTK-GNSS navigation, deployed on NVIDIA Jetson AGX Orin
- Designed a dual-EKF localization architecture fusing wheel odometry, 9-axis IMU, 2D LiDAR and dual-antenna RTK GNSS; implemented GNSS driver with NTRIP RTCM correction for centimetre-level positioning
- Built 3D obstacle avoidance from RGB-D camera using nvblox ESDF costmap layers, and tuned Nav2 with MPPI controller for real-world dynamic navigation

PPO Locomotion for Wheeled-Legged Quadruped

Aug 2025 – Oct 2025

Completed during internship at Immersion | nanj-robotics.github.io

SolidWorks · Isaac Gym · Reinforcement Learning · PPO · Robot Learning · Locomotion

- Modelled a 20-DoF custom wheeled-legged quadruped assembly in SolidWorks and exported the URDF; trained a PPO locomotion policy in Isaac Gym with 4,096 parallel GPU environments
- Designed a dense multi-term reward function with domain randomization (friction, mass, push disturbances, observation noise) and a flat→rough→stair curriculum for Sim2Real transfer
- Achieved 1.0 m/s forward speed and >95% velocity-tracking success on flat and rough terrain; converged after 5,000 iterations (~2.0 hours on RTX 4090)

ACADEMIC PROJECTS

FPGA Peak Detection Prototyping

Jan 2026 – Apr 2026

University of Bristol | VHDL · RTL Design · Digital System Design · FPGA

- Team-based hardware design project; implemented a signed-number peak-detection system on Xilinx Artix-7 FPGA using VHDL RTL coding style
- Co-developed the Command Processor module with an FSM controller and two-way handshaking protocol to synchronize high-speed FPGA logic with UART communication, ensuring zero data loss
- Team project grade: 94.1% (5-member team)

Fan-Duct & Heatsink Thermal Characterisation

Jan 2025 – Apr 2025

University of Bristol | Fusion 360 · 3D Printing · MATLAB · Python

- Worked in a team on an electromechanical cooling-system project covering mechanical design, numerical modelling, prototyping and experimental validation
- Completed 3D modelling of fan-duct hardware in Fusion 360 and prototype fabrication via 3D printing; developed parametric thermal modelling workflows in MATLAB/Python to predict heatsink thermal-resistance performance
- Participated in labs to characterise heatsink behaviour and measure LED efficiency; analysed experimental data and contributed to the final group technical report

TECHNICAL SKILLS

Programming	Python, C++, C, MATLAB, VHDL (FPGA design)
Robotics	Isaac Gym, ROS2, MoveIt2, ros2_control, Nav2, SLAM, AMCL, Pinocchio, RViz, Gazebo
Perception & ML	YOLO, FoundationPose, OpenCV, nvblox, PyTorch, LabelMe
Control & Estimation	PID, MIT-mode PD/torque control, cosine/S-curve trajectory generation, Pinocchio RNEA gravity compensation, EKF sensor fusion
Embedded & Hardware	CAN bus, LiDAR, RTK GNSS, IMU, RGB-D camera, NVIDIA Jetson AGX Orin, FPGA
Tools	Linux/Ubuntu, Git, conda, SolidWorks, Fusion 360

ADDITIONAL

Soft Skills: Team collaboration, technical communication, cross-functional coordination, project management

Languages: Mandarin Chinese (native); English (fluent — entire degree taught and examined in English)

Interests: Robot learning, robot perception, robot manipulation, mobile robots